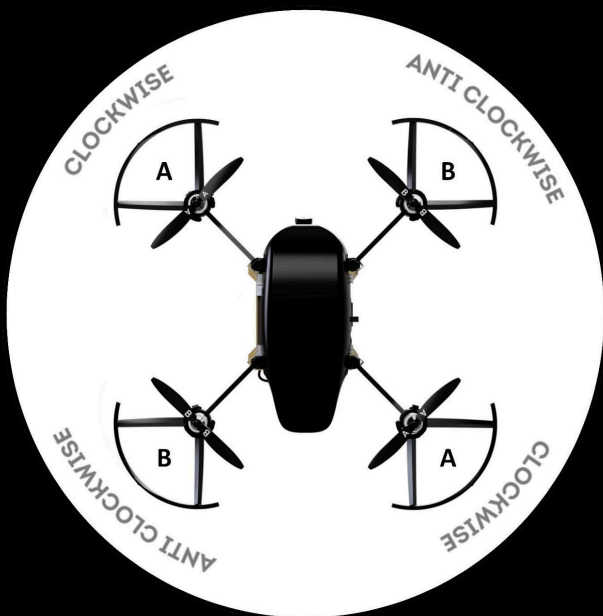




Pluto X User Manual



Fly in open areas



Maintain Line of Sight



Flying range - 60m



Avoid flying over or near obstacles, crowd, high voltage power lines, trees or bodies of water and near strong electromagnetic sources such as power lines and base stations.



Don't use Pluto X in adverse weather conditions such as rain, snow, storm etc. and in high wind



STAY AWAY FROM ROTATING PROPELLERS



DON'T FLY IN NO-FLY ZONE

Battery Safety

CAUTION: All instructions and warnings must be followed exactly. Mishandling of Li-Po batteries can result in fire, personal injury and/or property damage.

LI-PO BATTERIES: Failure to charge the battery with a compatible charger may cause a fire resulting in personal injury and/or property damage.

Rules and Regulations

Know About UAS Types

Nano	Micro	Small	Medium	Large
Up to 250 gm	More than 250 gm	More than 2 kg	More than 25 kg	More than 150 kg
	↓ Up to 2 kg	↓ Up to 25 kg	↓ Up to 150 kg	

According to the DGCA regulations, the Pluto X falls under the Nano drone category, as it weighs less than 250 grams. Here's how the rules apply to Pluto X:

Registration and Licensing

Nano drones are exempt from registration and do not require a Remote Pilot Certificate (RPC) for non-commercial use.

Operational Guidelines

1. **Altitude Limit:** Nano drones can be flown up to a maximum altitude of 50 feet (15 meters) above ground level.
2. **Visual Line of Sight (VLOS):** Operators must keep the drone within their direct visual line of sight during the flight.

3. **Time of Operation:** Nano drones can only be flown during daylight hours and in well-lit conditions.
4. **Restricted Areas:**
 - Flying over densely populated areas is prohibited.
 - Operation near airports, military installations, and other no-fly zones is strictly not allowed.
 - Operators should check the DGCA Digital Sky Platform for detailed airspace maps to ensure compliance.

For comprehensive information and updates on drone regulations, operators should refer to the official DGCA website and the Digital Sky Platform. Staying informed about the latest guidelines ensures compliance and promotes the safe integration of drones into Indian airspace.

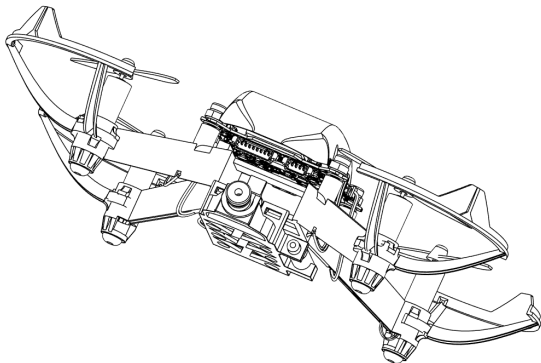
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Introduction

Hi!

We Know you must be excited to fly your Pluto X. Before you start tinkering on your own, we strongly recommend that you follow the quick start guide so that you have a seamless experience with your first flight. This manual also helps you assemble/disassemble your Pluto X and do more with your drone. Happy Tinkering!!



NOTE: Please read and understand this manual completely. Drona Aviation products are to be used by ages 8 and over. The instruction manual, warranties and other associated documentation is subject to change without notice. Drona Aviation assumes no responsibility for inadvertent errors in this manual.

Box Content

Pluto X is a ready to fly nano drone with Wi-Fi camera. The kit includes additional motors and spare parts to support hands-on learning and uninterrupted flying.

Content	Quantity
Assembled Pluto X Drone with Wi-Fi Camera	1
Extra Propellers (Set)	1
Propeller Removal Tool (Prop-tool)	1
Spare Motors	2
Pilot Badge	1
User Manual	1



Quick Flight

Step 1: Install “Pluto Controller” App Controller.



TIP:

Select “I am a Developer” while setting up the profile to enable user based programming. Visit the section “Developer options” for more details.

Step 2: Plug in Battery

A. Insert the battery from the front of the drone as shown in the figure.



B. Carefully route the battery connector and connect to the board connector.



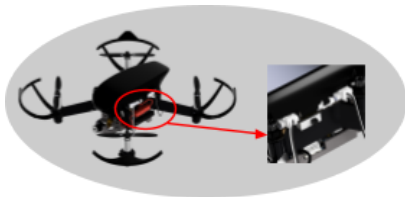
C. Use the connector holder to secure the connector to the frame.

CAUTION

The battery provided in the kit is charged to 20%. It is recommended that you charge the battery before your first flight.

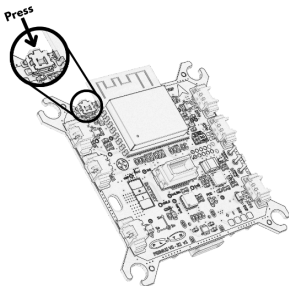
Step 3: Turn the Drone On

A. **Without Camera:** Switch ON Pluto X using the slide switch located on its right side.



B. **With Camera:** To use the camera, do the following:

1. Connect the camera using the connector provided in the box.
2. Before powering on the drone, **press and hold the boot button** located on the front side of the flight controller.
3. While holding the boot button, **switch ON** the drone using the slider switch on the side of the flight controller. Release the boot button immediately after the drone powers on.



4. Check the **ESP Blue status LED**, it will blink, which indicates that the camera has successfully booted.
5. Next, open the Pluto Controller application and tap on the Connect button. This will redirect you to your device's Wi-Fi settings.
6. From the Wi-Fi list, select **Pluto Cam_XXXX_XXXX** and connect.
7. After a successful connection, the camera is ready to use with your drone.

The camera can also be accessed from a laptop for image processing (feature under development).

Please Visit: <https://create.dronaaviation.com> for more updates.

Scan QR code to watch the video how to register and get your drone password.



NOTE: You must follow the same steps every time you access the camera and drone functions together.

Step 4: Connect to Your Drone Wi-Fi

To find the **password** for the **drone SSID**, refer to the instructions on the back of the cover.

- A. Open your Wi-Fi settings on your device.
- B. Locate the Pluto X_XXXX_XXXX SSID in the available networks list and select it.
- C. Enter the password to establish a connection.

Scan QR code to watch the video how to establish a drone connection.



Step 5: Preflight Check (Do this before every flight)

Scan the checklist, then follow the walkthrough if anything needs attention.

- ☐ Wires are not touching propellers
- ☐ Prop-guards secured
- ☐ Propellers 'A' and 'B' are properly configured
- ☐ Battery $\geq 3.8V$ before arming
- ☐ Motors on correct ports (M5–M8)
- ☐ Motor test passed (Spin All)
- ☐ Magnetometer calibrated
- ☐ Accelerometer calibrated

Preflight Walkthrough

A. Ensure motor wires are not touching propellers.

B. Propeller guards (prop-guards) are attached securely.

C. Propellers 'A' and 'B' are properly configured.



There are two sets of propellers A & B. You can find the markings near the hub of the propeller.

Make sure that the propellers are mounted as shown below:

'A' marked Propeller

'B' marked Propeller

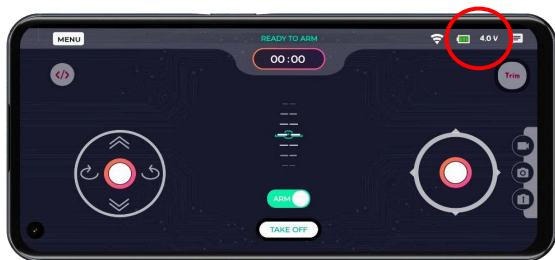
'B' marked Propeller

'A' marked Propeller



D. Battery check ($\geq 3.8V$ before arming)

Battery voltage before arming should be above 3.8V. This ensures good flight.



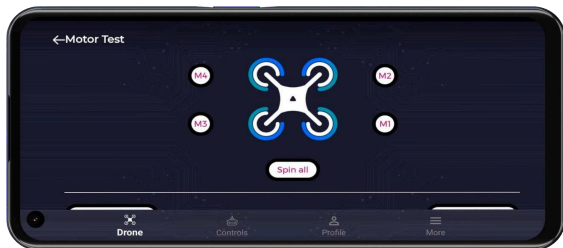
E. Motor ports: M5, M6, M7, M8

Please ensure that the motors are connected in the correct ports: M5, M6, M7, M8.

F. Motor Test (Pluto Controller App)

Perform motor test using “Pluto Controller”

- Go to Menu > Drone Settings > Motor test
- Tap on “**Spin all**”.
- Verify if all propellers are rotating. If not, please verify **step D**.



G. Magnetometer Calibration

This calibration accounts for stray magnetic fields. This is essential to achieve good yaw stability.

In the app, go to: Menu > Drone Settings > Mag. Calibration

TIP: It's a good practice to perform Magnetometer calibration when:

- A. Flying at a new location B. Recently replaced motors

CAUTION

Without proper magnetometer calibration, the yaw performance of the drone is severely affected.

H. Accelerometer Calibration

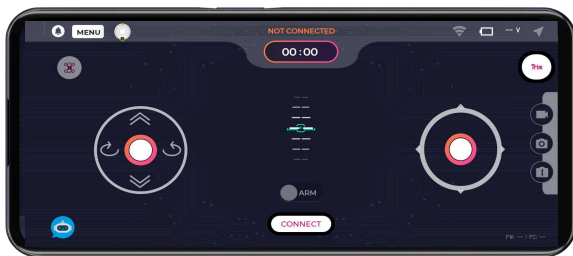
This calibration accounts for misalignment of the flight controller and frame.

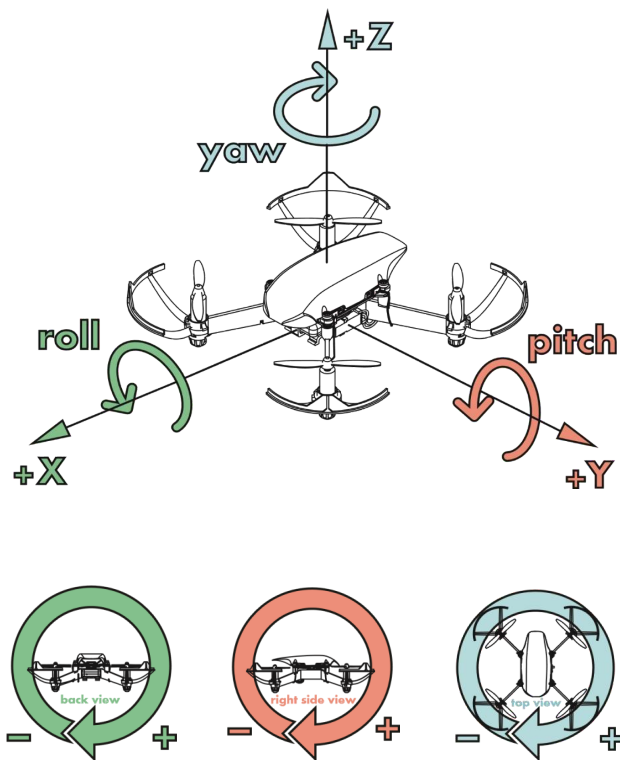


- Place Pluto X on a perfectly level surface.
- In the app, go to: Menu > Drone Settings > Acc. Calibration.
- This will reduce the drift, but not eliminate it completely.
- For eliminating the drift, you will have to trim the drone (please refer to section: Trim).

Step 6: Understanding controls

It's essential to understand the controls before your first flight. Please review the following:





Unlock your flying potential with a tap, watch our video tutorials for an unmatched flight experience!

To understand the
controls
Scan this QR code:



NOTE: If you are a beginner in flying drone, we recommend you to switch to “Tilt mode”. Go to Menu > Control Settings > Tilt mode

Step 7: Arming the Drone

The ARM switch controls power to the motors. It can be thought of as a safety switch.

NOTE: If the Arm button isn't active, return to calibration, some steps may have been missed.

Disarming the drone immediately stops all motors. Press the **Disarm** button to do so, but be cautious, using it mid-flight will make the drone drop instantly.

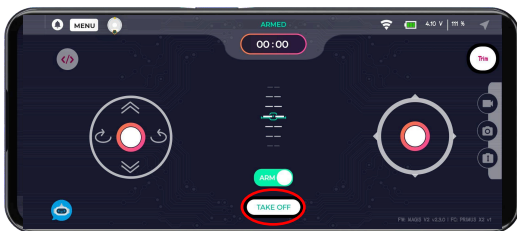
Step 8: Fly it!

- A. Place the drone with its front pointing away from you.

- B. Click the ARM button.



- C. Make sure you have a lot of open space.
- D. Press the Take-off button, Pluto X will automatically lift to a certain height/altitude.



- E. Focus on controlling only the drone's roll and pitch, this makes flying much easier.
- F. To land a drone, press the Land button.

NOTE: If you're new to flying, expect a few crashes while learning. Don't worry, Pluto X is crash resistant and built to handle minor impacts.

Battery Charging

Pluto X features an onboard battery charging system with a USB Type-C port, as shown below.

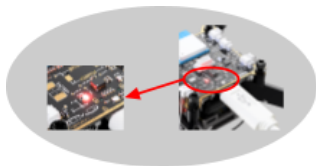
To charge the battery:

1. Turn Off the Pluto X drone.



2. Connect a C-type USB cable to the USB port.

3. Battery charging is indicated by a Red LED labeled “CHG”.



4. When the battery is completely charged, the Red LED stops glowing.

CAUTION

- Never discharge Li-Po Cells to below 3v under load.
- Never leave charging batteries unattended.
- Never cover warning labels with hook and loop strips.
- Never charge batteries in extremely hot or cold places.
- Never place batteries in direct sunlight.
- Never charge damaged batteries.
- Never allow minors to charge battery.

How To Replace Components

A. How To Replace Motor

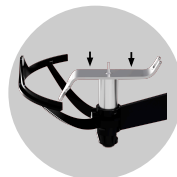
1. Remove the motors from the frame, use the propeller tool (prop-tool) to provide the right leverage.



2. Insert the motor connector through the end of the motor mount.

3. Place the motor in the motor mount.

4. Using a propeller removal tool (prop-tool), push the motors inside.



5. Harness the motor wires on the bottom side of the motor arm.

6. Connect the motor connector to the correct port (M5/M6/M7/M8).



B. How To Attach Canopy

1. To assemble the canopy, loosen the four flight controller screws.

2. The canopy has four mounts which go between the screw and the damper.

3. By flexing the canopy, wiggle in the canopy between the screws and dampers.

4. Tighten the screws if required.



NOTE: Loose screws can cause unstable flight, ensure all screws are properly tightened.

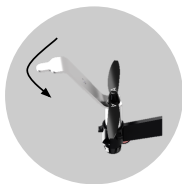
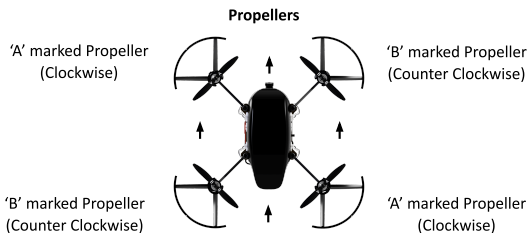
CAUTION

If the propellers are not installed according to the A and B configuration, the drone will fail to take off.

C. How To Replace Propellers and Prop-guard

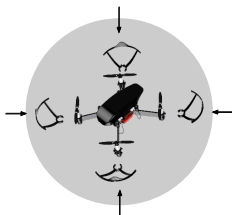
1. To replace, you need to take the propellers out, use the propeller tool (prop-tool) for proper leverage.

2. There are 2 sets of propellers labelled A & B. Please find the markings near the central hub of the propeller. Place the propellers according to the figure below.



3. If you need to take the propellers out, use the propeller tool (prop-tool) for proper leverage.

4. Slide in the propeller guards (prop-guards) as shown in the image.



CAUTION

If the propellers are not installed according to the A and B configuration, the drone will fail to take off.

D. How To Attach Battery

1. Insert the battery from the front of the drone as shown in the figure.



2. Carefully route the battery connector and connect to the board connector.

3. Use the connector holder to secure the connector to the frame.



Trim

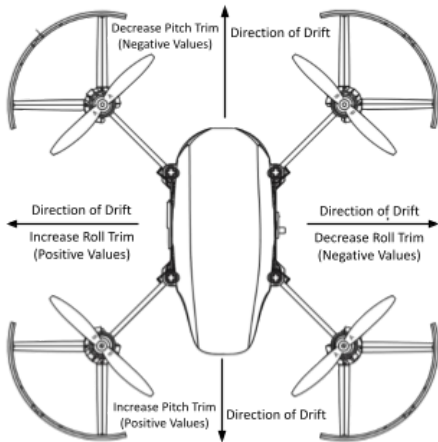
Trim is used to reduce minor drifts in your Pluto X. You may notice the drone moving slightly in one direction after take-off, trim feature helps stabilize your drone in the XY-plane.

Follow the steps given below to perform trim adjustment:

1. Make sure that you have performed accelerometer calibration before starting Trim.
2. Identify which direction is the drone naturally drifting towards.

3. Using Pluto Controller navigate to:
Menu > Drone
Settings > Trim

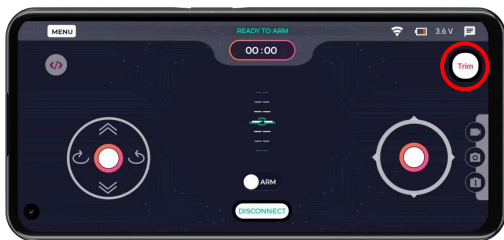
4. Trim the drone in steps of 5 for noticeable adjustment. Trim units are shown in the figure.



Inflight Trim

Trim is our new feature. The main purpose of this feature is to enable users to correct the drones drifting (left, right, front, back).

1. You can find the Trim at the top right of the main controller screen.



2. First, connect your Pluto drone. Once connected, tap on the **Trim** button. This will display two sliders around the Pitch and Roll joystick.



3. Use the **left slider** (up and down) to adjust the **Pitch trim** values. Similarly, use the **bottom slider** (left or right) to adjust the **Roll trim** values.

Control Settings

A. Control Modes define how you operate Pluto X, managing its movement (roll, pitch, yaw) and altitude (throttle/altitude hold).

The **Joystick** and **Tilt** modes specifically control **roll** and **pitch** behavior.

Joystick Mode	Tilt Mode
In this mode, the roll/pitch of the drone is controlled by an on screen joystick.	In this mode, the roll/pitch of the drone is controlled by tilting your phone. This mode is much easier to fly for beginners.

B. Flight Modes change how the Pluto X behaves during flight, particularly in the **Z-axis (up/down)** or orientation. Primarily altitude is controlled through **Throttle** or **AltHold** mode.

AltHold Mode	Throttle Mode
When the throttle control stick is centered, the drone maintains its altitude automatically. (Ideal for beginners for smoother flight.)	The drone's height changes with motor power, you must adjust continuously to maintain altitude. (Recommended for experienced pilots.)

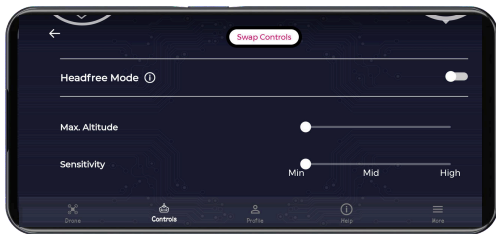
C. Headfree Mode affects how you manage the **roll and pitch** based on the drone's orientation.

Headfree On	Headfree Off
The drone automatically adjusts roll and pitch relative to mobile heading, no need to worry about yaw direction. <i>(Recommended for beginners.)</i>	You must track the drone's front direction (yaw). Losing orientation can make control challenging.

D. Sensitivity determines how quickly the drone responds to your control inputs.

Low Sensitivity	High Sensitivity
Larger input needed for noticeable change (slower, smoother response).	Small input = large movement (faster response).

Adjust this based on your flying style and comfort level.



NOTE: We recommend the beginners to keep the sensitivity levels to Low.

E. Maximum Altitude setting limits how high the drone can fly. By default, there is no altitude limit set. For safe indoor or controlled flights you can set one.

CAUTION

- This feature is still in beta, please use with care.
- This feature only works in Althold mode.

Update Firmware

Firmware is the software part of a drone, consisting of a sequence of commands that control its operations. It is generated and uploaded to the drone in the form of a hex file.

Steps using Wi-Fi:

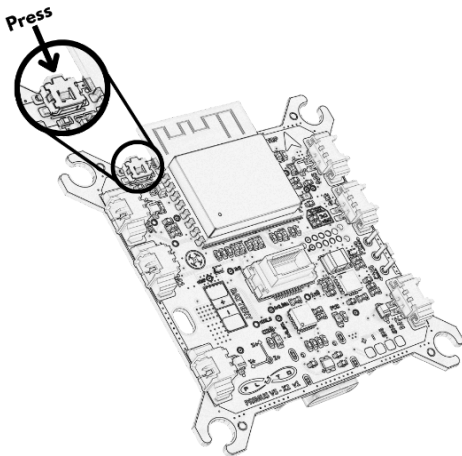
1. In the Pluto Controller app, navigate to **Menu > More > Update Firmware**.
2. Download the required firmware or select one from the **Browse** option. (Ensure the user is connected to the internet while downloading the firmware.)
3. Power on the drone and connect it to the Pluto Controller.
4. Choose the firmware file in “**.hex**” format and click on **Flash** to proceed.

Scan here or visit the link for firmware update assistance.



Steps Using Data Cable (USB A – C type or C – C Type):

1. Open the respective coding platform (VS Code - Pluto IDE).
2. Connect the flight controller to your system using the USB cable and Turn ON the Drone.
3. Press and hold the Boot button for about 5 seconds to enter Flashing Mode.
4. Click on Flash (or select Flash using USB) to upload the code.



More Ways To Fly Pluto X

Scan the below QR code to explore complete setup and flight tutorials.



1. Pluto Controller (Joystick)



2. Radio Controller (PPM Receiver & RC Transmitter)



3. External Controller



Developer Options

Pluto X is meant for tinkering. When a user wants to program the drone, use the Pluto IDE to write user code.

Along with Pluto IDE, enable developer mode on App. For more information on programming, use Project Book.

Please follow the steps to enable developer mode on App:

1. Go to **Menu** > Pilot Profile
2. Tap on the pencil option to edit your profile.
3. Tap on **"I am a developer"**
4. Go to the home screen of Pluto Controller app. **"</>"** button will be visible. Tap on this button to enable the user code.
5. Tap on the **"</>"** button to disable user code.



TIP: On Android devices, press the volume up button to enable the user code and volume down to disable it, instead of using the on-screen **</>** button.

Tinkering

Tinkering with your drone means exploring, customizing, and experimenting with its hardware or software to match your goals. Whether you're adding new features or enhancing performance, it lets you adapt the drone to your unique needs.

Scan QR code or visit link to
get start with Tinkering



Pluto Blocks provides an easy, block-based platform to program your drone's firmware. Its visual interface removes the complexity of traditional coding, making it simple for beginners yet powerful for advanced users. With Pluto Blocks, you can explore, experiment, and bring your drone ideas to life with ease.

For detailed instructions on using Pluto Blocks and
programming your drone firmware, refer to the
PlutoBlocks documentation on the website.





Pluto **IDE**

Pluto-IDE is a Visual Studio Code extension designed for smooth firmware development and deployment on the Pluto Drone. It simplifies the entire workflow, from coding to flashing, offering an integrated environment that removes the hassles of traditional embedded toolchains.

(For detailed instructions on using Pluto IDE and programming your drone firmware, refer to the Pluto IDE documentation on the website.)



Pluto can be programmed using Python, a simple and versatile language. With the Pluto Python wrapper, you can write custom scripts to control your drone's movements and create your own projects using predefined functions.

(For detailed instructions on using Python and programming your drone firmware, refer to the python documentation on the website.)



ROS

Pluto can be operated using ROS. ROS stands for Robot Operating System. It is a set of open-source libraries which help software development of robot applications.

(For detailed instructions on using ROS and programming your drone firmware, refer to the ROS documentation on the website.)



Troubleshooting

Led Status

For reference: R - Red LED, G - Green LED, B - Blue LED

LED Status	Indication	Action required
Cyclic R, G, B	Waiting for the app to connect.	Connect using App
G On	Connected with App. Ready to ARM	-
B On	Drone is Armed	-
R On	Low battery	Charge the battery
R Toggle	Pluto X is not of a flat surface	Keep Pluto X on a flat surface
B Toggle	Signal Loss	Reconnect to Pluto X

Activity LED (Small Blue LED)

Mode	What It Means	LED Behaviour
FW Flash Mode	Putting Drone into Flashing Mode	LED blinks very fast
Controller Activity	Drone is sending or receiving data from Pluto Controller (Wi-Fi)	LED blinks quickly
Wi-Fi Connected	Drone's Wi-Fi is connected to your device	LED blinks slowly (once per sec)
Wi-Fi On (No Device)	Drone's Wi-Fi is on, but no device is connected	LED stays ON (no blinking)
Camera Mode	Drone is in camera mode	LED blinks briefly every 2 sec

Failsafe Information

Failsafe	Condition	LED	Behaviour
Low Battery (Not armed)	Batt voltage < 3.3V	Shows Red LED	Does not let you ARM the drone
Low Battery (In-flight)	Batt voltage < 3.0V	Blink Red LED	-
Loss of Range	No signal for 2s	Blink Blue LED	Land drone at the current position
Crash	Acceleration > 2g or Roll/Pitch > 50 degrees	Blink Red and Green LED	Disarms the drone

FAQs

Q.1 Why are some motors not rotating?

This can happen for several reasons. Please check the following:

- Ensure each motor is connected to the correct ports (M5, M6, M7, M8).
- Inspect motor cables for any damage; replace the motor if necessary.
- Confirm that the propeller rotates freely on the motor shaft. If it's tight, use the prop-tool to create a small gap between the motor and the propeller hub.

Q.2 Why is the drone not taking off?

Please check the following:

- Ensure the battery is fully charged.
- Verify that the propellers are mounted correctly (refer to the *Assembly/Disassembly* section).
- Confirm that your device is connected to the Pluto X Wi-Fi network.
- After connecting through the app, check for a solid green LED on the drone.

If the LED is not solid green, refer to the *LED Status* section to identify and fix the issue.

Q.3 Why does the drone keep drifting?

Some drift is normal since the drone controls angles, not velocities. To minimize it, try the following:

- Perform Accelerometer Calibration on a flat, level surface.
- Follow the steps in the Trim section to further reduce drift.

Q.4 Why does Altitude Hold behave erratically?

Pluto X maintains altitude using a MEMS barometer, so a variation of ± 50 cm is normal. If the drone shows excessive vertical movement, it may be due to vibration.

Please check the following:

- Replace any damaged propellers, as even minor damage can cause high vibration.
- Ensure the Primus X board touches only the dampers and not the plastic frame.

Q.5 Why can't I see the Wi-Fi hotspot?

Try to restart the drone and check again in your device's Wi-Fi list.

Q.6 Why does the drone keep rotating or yawing?

Make sure the magnetic calibration has been performed correctly.

Q.7 What is the maximum payload capacity of Pluto X?

Pluto X can carry up to 10 grams.

Q.8 How far can Pluto X fly?

The drone has a range of up to 100 m, but for optimal visibility and control, it's recommended to fly within 40 m.

Q.9 Where are photos and videos from the drone camera saved?

If an SD card is inserted, files are saved there. Otherwise, they're stored locally on your device and can be accessed via your phone's Gallery or Photos app.

Q.10 Where can I buy spares for Pluto X?

All spares are available on our official store:

<https://www.dronaaviation.com/store/>

Q.11 Is it legal to fly Pluto X in India?

Yes. Pluto X is a nano drone (<250 g) and is exempt from DGCA restrictions or licensing requirements.

Refer to the official notification:

<https://egazette.gov.in/WriteReadData/2021/229221.pdf>

Q.12 How can I control Pluto X using my phone's motion sensors?

Go to Drone Settings → Tilt Mode in the app to enable motion-based flight control.

Q.13 How do I perform a flip with Pluto X?

While flying, press and hold the Flip/Aux button, then move the direction stick toward the side you want to flip.

Specifications

Drone Specifications

Height	50 mm
Frame Size	160 mm
Clock	72 MHz
Wireless Communications	Wi-Fi Interface
Propulsion	Brushed Coreless Motor
Propeller	Bi-Blade 55 mm
Weight with Battery	60 g
Maximum Range	Up to 80 m

Battery Specifications

Voltage	3.7 V
Rated Capacity	600 mAH
Discharge Rating	25 C
Chemistry	Lithium Polymer

Primus X2 Flight Controller Specifications

Microcontroller	STM32F303CB32-Bit ARM Cortex M4 Frequency: 72MHz Flash: 256kb
Sensors	INA219: Voltage and Current ICM-20948: 9-axis IMU: Accelerometer, Gyrometer, Magnetometer ICP-10111: Barometer, Pressure, Temperature
Internal Power	V_{in} : 2.8v - 5.5v Current rating: 0A - 2A ESD safe Logic level: 3.3v on-board 1s Battery Charger: 0.9A to 1A
Motor Drive	4 x N-Channel MOSFET (6A) 2 x DRV8835: H-Bridge drives (3A)
Wi-Fi	Access Point, Station mode
On Board	Camera, SPI, I2C, PPM 5 x ADC, 3 x PWM, 6 x GPIO
Uni Bus	UART2, SPI, I2C, 8 x ADC, 6 x PWM, 14 x GPIO
Physical	Length: 48mm, Width: 48mm, Mounting Hole (Dia): 3.6mm, Weight: 10.4g
Programming	AeroFlash OTAUSB DFU Mode

Camera Specifications

Photo Format	JPEG
Picture Resolution	1080 ppi
Video Resolution	1280p x 720p
Video Standard	High Definition
Camera Sensor	CMOS
Storage	Phone Internal Storage/ SD Card Storage



Scan here to start your
drone tinkering Journey!



Scan here to watch
video tutorials!

For questions and technical help:

Email: support@plutodrones.com

Call: **+91-7066011808**

GitHub: <https://github.com/DronaAviation>

Forum: <https://www.dronaaviation.com/forum>

REGISTER YOUR PLUTO DRONE

Register Your **Pluto Drone** Today to get
your Password & activate **Your Warranty**



**Scan Here For
Video Tutorial**

SSID and Auth Key



Auth Key is not the password



Step 1: Open Menu
in Pluto Controller



**Step 2: Go To
Profile in Menu**



**Step 3: Switch To
The Hangar Tab**



**Step 4: Tap Add a
New Drone**



**Step 5: Fill In
Drone Details**



**Step 6: View The
Password in Hangar**